

Underwater Vision-Based Surveillance Using ML and AI

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12 — Dataset V1 Review and Finalization

1. Objective

To review EDA findings, annotations, preprocessing, augmentation and split logic, resolve identified issues, and prepare Dataset V1 for model training.

2. Dataset Summary

Dataset	Classes	Train	Valid	Test
DIATAquarium	11	8,121	773	Not available locally
Aquatic Plant	1	1,892	179	89
Well	4	3,795	191	184

Total available images: 15,224

3. Problems Identified

DIATAquarium

- 11 classes.
- 657 empty label files.
- 0 invalid annotation entries.
- Local test folder is not available.
- data.yaml points both val and test to test/images.

Aquatic Plant

- 1 class.
- 13 training images without matching labels.
- 1010 empty label files.
- 0 invalid annotation entries.
- 6 duplicate images found in the checked training sample.

Well

- 4 classes.
- 15 training images without matching labels.
- 39 empty label files.
- 0 invalid annotation entries.
- Severe class imbalance — Fishes: 50,194, School-of-fish: 653, Stone: 5, Inlet-pipe: 3.

4. Annotation and Image Quality Review

- YOLO annotation format was verified.
- 0 invalid annotation entries were found across all three datasets.
- Missing-label images were reviewed; some appeared to be valid background/no-object images.
- Empty labels were therefore retained rather than automatically deleted.
- Representative samples showed haze, low visibility, low contrast and colour variation.

Sampled resolutions:

- DIATAquarium Train: 1920 × 1080
- DIATAquarium Valid: 2048 × 1536
- Aquatic Plant: 640 × 640
- Well: 640 × 640

5. Preprocessing Finalized

Original Image → Resize 640×640 → RGB → Normalize 0–1 → Model Input

Reasons:

- 640×640: provides a consistent model input size across datasets.
- 0–1 normalization: provides a consistent pixel range.
- RGB: provides consistent colour-channel representation.
- Raw data: remains unchanged.

6. Enhancement and Augmentation

Classical Enhancement

White Balance + CLAHE

- White Balance → reduces colour distortion.
- CLAHE → improves local contrast.

Training Augmentation

- Horizontal flip
- Small rotation
- Scale variation
- Brightness/contrast variation

Augmentation will be applied only to training data. Validation and test data remain unchanged for fair evaluation.

7. Source-Level and Split Analysis

Source-level analysis was performed because many images originate from video recordings. Randomly splitting individual frames from the same recording can place very similar frames into different splits and cause data leakage.

Source Count

Dataset	Number of Sources
DIATAquarium	6
Aquatic Plant	2
Well	1

DIATAquarium Sources

The 6 identified sources are:

- av0_0_20231228_094444_mp4
- av0_0_20231228_102111_mp4
- av0_0_20231228_103918_mp4
- av0_0_20240222_102152
- av0_0_20240222_103703
- av0_0_20240222_105838

All 6 sources appear in both the train and validation splits.

The source-level class analysis also showed that classes are not evenly represented across sources. For example:

- 102152 contains all 11 classes.
- 103918 contains 7 classes.
- 094444 contains 5 classes.

Therefore, source-based re-splitting must consider class coverage, not just source count.

Aquatic Plant Sources

Two sources were identified:

Source	Train	Valid	Test
2023-12-09_13_18_14_mp4	585	61	26
2023-12-09_13_20_33_mp4	1,294	118	63

Both sources appear across the three splits.

Well Source

Only one source was identified:

Source	Train	Valid	Test
20240102-20240102153559-20240102153818-153600_mp4	3,780	191	184

The same source therefore appears across all three splits.

Split Decision

A fully source-independent split is not possible for all datasets because:

- Aquatic Plant has only 2 independent sources.
- Well has only 1 independent source.

Therefore, randomly moving individual frames between train, validation and test is not used as a solution.

Decision: The existing supplied train/validation/test structure is preserved for Dataset V1. Source-level leakage is documented as a limitation and will be considered when interpreting final model results.

Exact cross-split duplicate checking found 0 duplicates in the checked samples. The remaining concern is therefore near-identical frames from the same source, rather than identical files.

8. Dataset V1 Final Configuration

Parameter	Final Setting
Datasets	3
Classes	16
Images	15,224
Image size	640 × 640
Normalization	0–1

Parameter	Final Setting
Colour format	RGB
Annotation	YOLO
Augmentation	Training only
Raw dataset	Unchanged
Invalid annotations	0

9. Final Decision

The following preparation strategy is finalized for Dataset V1:

Raw Dataset → Quality Check → Existing Split Preservation → Resize 640×640 → RGB → Normalize 0–1 → Training Augmentation → YOLO Training

The original datasets remain untouched, and Dataset V1 is maintained as a separate versioned copy.

10. Mentor Review

The following points will be confirmed during mentor review:

- Dataset split strategy
- DIATAquarium test-set handling
- Source-level leakage limitation
- Preprocessing and augmentation settings

Any mentor corrections will be incorporated before the final freeze.